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Personality, affective response, and facial blood flow during brief cognitive tasks

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Abstract

Blood pressure, heart rate, and changes in facial and finger blood flow were monitored in 58 women during three laboratory tasks, i.e., reading out a neutral text, a personally relevant speech (including a silent preparation phase), and a tracking task. Participants rated the tasks as mild to moderate with regard to affect intensity. Significant effects of personality on affect states were demonstrated, indicating higher levels of anxiety, depression, and anger in the high-neuroticism group, and more curiosity in the high-extraversion group. Arterial pressure, heart rate, and facial blood flow increased during task performance in all three conditions. No relationship between facial blood flow changes and muscle activity measured by m. masseter electromyography (EMG) were demonstrated. Digital vasoconstriction occurred in parallel with facial vasodilatation during tracking and in the preparation phase before the speech. There were no moderator effects of neuroticism or extraversion on blood flow or other cardiovascular responses.
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Keywords: Facial blood flow; Cardiovascular responsiveness; Affect state; Neuroticism; Extraversion

1. Introduction

Although considered a fairly common phenomenon in human social interaction, emotion-linked facial blood flow changes have not been extensively studied using psychophysiological methods. The scant research so far has shown that facial blood flow tends to increase during anxiety-provoking tasks, such as mental arithmetic and difficult puzzles, or in response to anger-inducing provocation (Drummond, 1994, 1999; Drummond and Quah, 2001; Nordin, 1990). Importantly, individual differences in how readily facial blood vessels dilate during different stressful stimuli might be a contributing factor in facial flushing or blushing propensity (Drummond, 1997b). A further issue, which has received almost no attention, is whether personality traits are related to facial blood flow changes during emotion-inducing situations and tasks. The primary purpose of the present study was to

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investigate facial blood flow as part of cardiovascular response patterns during cognitive tasks, and to determine the effects of personality variables on physiological responding and affect states.

Accumulated research indicates that neuroticism and extraversion are important determinants of individual variability in emotional responsiveness in everyday life (Costa and McCrae, 1980, 1992; Eysenck, 1990). Generally, high-neuroticism individuals have a tendency to experience more negative emotions such as fear, sadness, embarrassment, anger, and guilt than low-neuroticism individuals do (Watson and Clark, 1984). Individuals who exhibit a high degree of extraversion tend to experience more positive emotions such as joy, happiness, and excitement in everyday life than less extraverted individuals do (Costa and McCrae, 1980, 1992). Experimental manipulation studies have likewise documented moderate to strong personality–affect relations (Gross et al., 1998; Schwebel and Suls, 1999; Watson and Clark, 1984). Furthermore, Gross et al. (1998) showed that correlations between extraversion and positive affect, although significant, tended to be weaker than personality–affect relations in the domain of negative affect. In a recent study using functional magnetic resonance imaging, Canli et al. (2001) demonstrated that extraversion was correlated with brain reactivity to positive stimuli in localized brain regions, and neuroticism was correlated with brain reactivity to negative stimuli in localized brain regions.

Thus, it should be expected that both these personality dispositions have an impact on affective response during challenging cognitive tasks and, consequently, on changes in facial blood flow. To our knowledge, only one study has addressed this question. Drummond (1994) explored the relationship between personality traits and changes in heart rate and facial and finger blood flow during mental arithmetic. Neuroticism, extraversion, and psychoticism (Eysenck and Eysenck, 1975) did not predict the size of the cardiovascular response. However, effects of personality traits on the *pattern* of physiological responding during the task, i.e., moderator effects or interactions, were not explored. In a later study, Drummond (1999) showed that, whereas subjects high in trait anger reported most anger and embarrassment during a difficult puzzle, physiological responses, including facial blood flow changes, did not differ from subjects with low trait anger.

Possibly, the lack of significant findings may be a consequence of the cognitive tasks (i.e., mental arithmetic, puzzle) used in these studies. A growing body of research has demonstrated that social and self-involving situations or tasks, such as provocation, conflict, or self-evaluative threat, elicit substantial changes in cardiovascular responses and other physiological endpoints (e.g., Bongard et al., 1997). Moreover, individual differences in socially relevant personality dimensions appear to moderate physiological reactivity to social stressors (e.g., Gallo et al., 2000). Thus, in studies of personality–affect relations, the choice of personality-relevant stimuli is mandatory.

In one of their studies, Drummond and Quah (2001) assessed cardiovascular responses to a supposedly nonstressing comparison task. In this study, subjects were required, among other things, to describe anger-provoking incidents or to read out neutral material (i.e., gardening passages) either loudly and rapidly or softly and slowly. Both forehead and finger pulse amplitude changed significantly during anger expression, but not when reading out neutral material. These results are in congruence with previous research indicating that speech content (i.e., affect-related vs. neutral) has significant impact on cardiovascular responding during speech (Linden, 1987; Tardy and Allen, 1998). Linden (1987) concluded that motor efforts required for speech production are of negligible importance in predicting general cardiovascular responding (e.g., blood pressure and heart rate) to tasks involving speech. Possible associations between local facial blood flow changes and muscle activity have not been investigated, however. Thus, research is needed to determine facial blood flow responses and corresponding changes in muscle activity across tasks varying in terms of affect intensity and response mode (e.g., verbal vs. nonverbal).

For the most part, changes in forehead blood flow have been monitored during the experimental tasks. Only a couple of studies have also included measures of cheek blood flow changes. Drummond (1994) demonstrated that skin blood flow increased in the forehead and cheek but decreased in the finger during mental arithmetic. Moreover, cheek pulse amplitude (but not forehead pulse amplitude) increased in a reward condition. In another study (Drummond and

Lance, 1987), both forehead and cheek blood flow increased during a supposedly embarrassing task.

Accumulated research indicates that there may be gender differences in the reported intensity of emotional experiences (Shields, 1991), as well as in the cardiovascular response to acute stress (e.g., Allen et al., 1993; Girdler et al., 1990). Moreover, women show more physiological reactivity in accordance with emotion valence judgments than men do (Lang et al., 1993). Finally, gender has been found to moderate associations of blood pressure with personality measures (Jorgensen et al., 1996). Thus, acute stress can trigger different patterns of affective and psychophysiological reactions in women and men, with women, on the average, appearing prone to high levels of stressor-induced emotional distress. Consequently, the use of a female sample was expected to provide a sensitive test of associations of neuroticism with the psychophysiological and affective measures obtained during the social stressor.

In the present study, the relationships between personality, affect states, cheek blood flow changes, and general cardiovascular responsiveness were examined in a sample of healthy women. Specifically, physiological and affective responses were assessed in one experimental session comprising three cognitive tasks (i.e., reading a neutral text aloud, a personally relevant speech task, and visual tracking), and modifying influences of personality were analyzed. The laboratory tasks were selected both because of their frequent use in cardiovascular stress research (Schneiderman et al., 2000) and because of their dissimilarity regarding both type of challenge and the coping response required of the subjects. Thus, the experimental manipulations included both verbal and nonverbal tasks, and the two verbal tasks were highly different in terms of personal relevance and social-evaluative threat potential. Moreover, the speech task included a planning phase in which the subjects were asked to silently prepare themselves for the speech, focusing on predetermined topics. Finally, to investigate effects of jaw-muscle activity on cheek blood flow, the participants were told to bite on a masseter muscle ergometer.

It was hypothesized that facial blood flow would generally increase during active task performance, with the strongest response occurring in the speech task. Furthermore, it was anticipated that extraversion and neuroticism would be associated with positive and negative emotional experiences, respectively, throughout the experimental session. Hence, it was expected that high neuroticism subjects would also exhibit stronger facial flushing, particularly during the personally relevant speech task. The effects of positive emotions and extraversion on cardiovascular responding were more uncertain, however.

2. Method

2.1. Subjects

Fifty-eight healthy women with a mean age of 35.9 years (S.D.=7.8) were included in the study. The participants were recruited from the central administration of the University of Oslo, from graduate classes, and among employers at an agency providing secretarial substitutes. Mean level of education was 7.4 years (S.D.=1.4) beyond 9-year compulsory schooling. Exclusion criteria were chronic medical illnesses, regular use of medication other than oral contraception, and psychopathology, as indicated by self-reported psychiatric diagnosis, or a high score on the Symptom Checklist 90 Revised (SCL-90-R), i.e., a mean score above 2.30 (see Derogatis, 1983; Vassend et al., 1992; Vassend and Skrondal, 1999). Originally, 60 subjects were tested. However, two subjects were later excluded; one due to self-reported psychiatric diagnosis and the other due to elevated distress, as indicated by the SCL-90-R score. All subjects received written information of the investigation, and all signed a consent form approved by the regional Ethics committee. The subjects were paid NOK 500 (about US$65) for their participation.

2.2. Self-report instruments

Neuroticism and extraversion were measured using the Revised NEO Personality Inventory (NEO PI-R; Costa and McCrae, 1992). The inventory consists of 240 items measuring five broad domain factors (i.e., neuroticism, extraversion, agreeableness, conscientiousness, and openness). Several studies of the Norwegian version of the NEO PI-R have indicated satisfactory reliability coefficients and very high coefficients of congruence for the five domains when

compared with the U.S. normative sample (see Kornør and Nordvik, 2004).

The subjects' affective state *during* the experimental tasks was assessed retrospectively using 10 Visual Analogue Scales (VAS). Immediately after each task, the participants were asked to rate the intensity of their emotion during the task (e.g., feeling irritated, tense, satisfied, and relaxed) from "not at all" (0) to "maximally" (100). This method was used to obtain brief ratings of positive and negative affects linked explicitly to the participants' perception of the tasks and their own task performance.

Affect states *between* tasks were assessed using the state part of the State Trait Personality Inventory (STPI; Håseth and Spielberger, 2000; Spielberger et al., 1995). The participants were asked to rate their affect state at the moment, "right now". The STPI measures state and trait anxiety, depression, curiosity, and anger. Each scale consists of 10 items rated on a four-point scale. Unpublished results from a psychometric study (N=294) of the Norwegian STPI version have indicated satisfactory alpha reliability coefficients for the four state measures, i.e., 0.62, 0.83, 0.93, and 0.96 for curiosity, depression, anxiety, and anger, respectively (Håseth and Spielberger, 2000).

Essential health behaviors were measured using the following indices: (1) frequency of physical exercise (moderate to high-intensity exercise, causing breathlessness and sweating, for 20 min or more), rated on a five-point scale ranging from 0 (never/very seldom exercise) to 4 (exercise four times or more per week); (2) consumption of alcohol, rated on a six-point scale ranging from 0 (never/very seldom drink alcohol) to 6 (drink alcohol four times a week or more); (3) smoking, scored 1 (regular smoker) and 0 (do not smoke); and (4) body mass index (BMI), calculated using the formula $BMI=weight/(height)^2=kg/m^2$ (Stensland and Margolis, 1990).

2.3. Physiological data collection

Mean arterial blood pressure (MAP) and heart rate (HR) were monitored by the Penãz method, using an Ohmeda Finapres 2300 (Englewood, CO, USA) monitor. The Finapres provides continuous recording and has been shown to yield high reliabilities for change scores, as well as absolute levels of MAP and HR. A pressure cuff, servo controlled by a light emitting diode and sensor, was attached to the middle phalanx of the third finger on the subject's left hand.

Skin blood flow changes were recorded with a Perimed Multichannel Laser Doppler System (Periflux 4001 Master, Perimed, Sweden), with miniature probes (Perimed) attached to the left m. masseter area and the ventral side of the left thumb. This instrument measures skin blood flow in arbitrary units of flux, proportional to the velocity and concentration of red blood cells moving in the superficial layer (i.e., 1–2 mm below surface) of the skin. Thus, this technique does not give absolute values but permits comparable measurements of relative changes in the skin blood flow across subjects (Johnsson et al., 1984). In the analysis of within-subjects changes, laser Doppler flowmetry (LDF) measures expressed in arbitrary units can be employed.

Muscle activity was recorded with surface electromyography (EMG) from the left m. masseter using custom-made electrode assemblies with built-in preamplifiers. Electrode diameter was 5 mm, and the interelectrode distance was fixed at 20 mm. The skin was cleansed with alcohol, and Ag/AgCl electrode paste was applied to the electrodes.

All signals were AD converted, recorded, stored, and reduced in a LabView system (National Instruments, Austin, TX, USA). The AD sample rate was 2 kHz. All physiological variables were averaged for the entire time period for each of the experimental phases (i.e., baseline, task, recovery, and isometric contraction, see below).

2.4. Tasks and procedures

The present study was part of a larger project also including patients with temporomandibular pain problems. Thus, in addition to the measurements reported here, an assessment of pain sensitivity (using pressure pain and electrocutaneous pain stimulation) was performed, and additional paper and pencil tests were administered after the laboratory tasks. Moreover, muscle activity was recorded with surface electromyography from the left m. biceps brachii and bilaterally from the mm. trapezius during the tasks, as well as during isometric jaw-muscle contraction and maximal voluntary contraction of arm and shoulder muscles by the end of the experimental session. These data fall outside the aims and scope of

this paper and will therefore not be presented here. Only healthy controls from the larger project were included in the present analyses.

The subjects were required to refrain from vigorous physical activity, drinking tea or coffee, smoking, and having large meals during the last 4 h before the laboratory session started. Alcohol was not allowed the last 12 h before the experiment. The subjects first went through the laboratory session, lasting about 2.5 h, after which they completed the NEO PI-R and other self-report instruments. It was considered particularly important to familiarize subjects with the laboratory procedures to reduce apprehensive anticipation and increased levels of activation (Fahrenberg et al., 1995). Upon arrival at the laboratory, each participant was given a brief explanation of the experiment, and informed consent was obtained. Next, the subject was seated in an upright position in a sound-attenuated and electromagnetically shielded room. The sensors were attached, and rating scales and other procedures were demonstrated.

The experimental sequence was as follows: (1) STPI, (2) reading task followed by VAS ratings, (3) speech task and VAS, (4) STPI, (5) tracking task and VAS, (6) isometric contraction, and (7) STPI.

2.4.1. Reading aloud

Originally, this task was intended primarily to promote habituation to the experimental situation. Thus, it is somewhat briefer than the following tasks. In the present context, it is included also to assess cardiovascular and emotional responses to a familiar verbal activity. Subjects were first instructed to relax for 1.5 min. Then they were asked to read aloud for 1.5 min from a chapter in a book of the history of Australia. The history text was intended as a neutral (i.e., not personally relevant) topic. Next, the subjects again relaxed for 1.5 min, after which the affect scales (VAS) were completed.

2.4.2. Speech task

The personally relevant speech-stress task was initiated by a 3-min period of relaxation. Then, subjects were instructed to prepare themselves for a job interview, concentrating on professional and personal skills, achievements, etc. They were told to plan their speech first and start talking when the experimenter asked them to. Both the planning and the speaking phases lasted 3 min. After finishing speaking, subjects again relaxed for 3 min. Finally, the VAS ratings were completed.

2.4.3. Tracking task

The tracking task was initiated by a 3-min relaxation. Then, a black square started moving across the computer screen in front of the subjects, who were instructed to track the cursor for 3 min, using the computer mouse. After 1.5 min, while the subject was still performing the tracking task, electrocutaneous pain sensitivity was tested. Thus, in the present context, the recordings obtained during the 1.5-min period prior to pain stimulation were used in the analyses. After tracking, the subjects were again instructed to relax for 3 min. Finally, the VAS scales were completed.

The experimenter (a female research fellow) was present in the same room with the subject throughout the laboratory session. Besides practical considerations, this arrangement was chosen with the purpose of eliciting interpersonal processes and feelings in the experimental situation, thus increasing the emotional impact of the tasks.

The fixed sequence of presentation of the cognitive tasks may, of course, produce "carry-over" effects of cardiovascular responding between tasks. However, it should be emphasized that the primary aims of the study are to characterize cardiovascular response *patterns* during task performance and to determine the *relationship* between personality traits and physiological responding. Thus, while individual cardiovascular response amplitudes might have been different using other task orders, there is no reason to believe that the direction of the response, such as the uniform facial vasodilatory response observed during task performance (see Results section below), would have been markedly different. Moreover, to investigate the relationship between personality variables and cardiovascular responding, the possible role of task order and the precise characterization of individual response amplitudes are of no major concern. Finally, the lack of differences between baseline and recovery response levels across the experimental session (see Results section below) indicates that carry-over effects were negligible.

2.4.4. Isometric contraction

Subjects were instructed to bite on a custom-made muscle ergometer for 1 min using constant force. To ensure that the subjects maintained the predetermined pressure level during the 1-min period, an oscilloscope provided visual feedback of the pressure exerted on the ergometer.

2.5. Data analysis

For each baseline, task, and recovery period, means and standard deviations were taken of MAP (mm Hg), HR (beats/min), and blood flow (laser Doppler flowmetry; LDF) data. LDF data were analyzed using change scores, computed as a percent change from baseline to task in each condition. Thus, for each task, the main effects of Group (i.e., high/low neuroticism or extraversion), Phase within task (i.e., baseline, task performance, and recovery), and Group×Phase interaction effects were analyzed using two-way multivariate repeated measures analysis of variance (MANOVA). Significant effects were further examined using *t*-tests for comparisons of means. However, when LDF change scores are used, differences in physiological response levels between baseline and task performance cannot be tested because the baseline value is a constant (i.e., set to 100 in the present study). Thus, to analyze within-subjects changes from baseline to task, LDF raw scores (expressed in arbitrary units) were used. A probability value (p) of less than 0.05 (two tailed) was considered to represent a significant effect.

It can be argued that the median split procedure reduces the likelihood of identifying influences of personality on psychophysiological responses because group membership near the middle of the distribution is rather arbitrary. Thus, correlation and regression analysis were conducted to investigate associations between personality traits and affect states, on the one hand, and physiological responding during task performance, on the other. Using these statistical procedures, the information contained in the entire range of each measure will be utilized. However, the pattern of responding across phases within or between tasks cannot be analyzed with these methods.

Finally, the relationship between EMG response (relative to baseline) and blood flow changes was examined during the cognitive tasks, as well as during isometric jaw-muscle contraction. To compute changes in EMG and LDF during the contraction condition, the recovery period after tracking was used as baseline. All values in figures are given as change scores±S.E.M.

With the sample size used in this study (although somewhat larger than what is common in this research area), results may be unduly influenced by extreme scorers. However, the main results from the analyses did not differ substantially whether statistical outliers, identified on the basis of box plots or scattergrams, were included or excluded. Mean values and S.E.M. indicated in the figures were computed with statistical outliers excluded (0–2 per figure). Thus, the number of subjects varies somewhat across analyses.

3. Results

3.1. Self-report measures

Neuroticism scores ranged from 27 to 129 (mean=83.98, S.D.=18.68), and extraversion scores from 73 to 147 (mean=117.70, S.D.=16.51). These scores are within the average range of the original American normative sample (Costa and McCrae, 1992). The mean neuroticism scores in the high- and low-neuroticism subgroups were 98.44 (S.D.=12.60) and 70.03 (S.D.=11.59), respectively. For the high- and low-extraversion subgroups, the mean extraversion scores were 130.90 (S.D.=8.10) and 105.81 (S.D.=11.62), respectively.

The participants reported higher levels of negative affect, particularly tension and embarrassment, and lower positive affect levels in response to the speech task as compared with the other conditions (Table 1). However, for all tasks, the positive affect ratings were higher than the negative affect ratings.

While neuroticism was related to negative affect ratings in expected ways, the correlation coefficients reached significance only in the speech and tracking conditions. Thus, neuroticism was associated with "tense" [$r(57)=0.35$, $p<0.01$], "depressed" [$r(57)=0.42$. $p=0.001$], and "apprehensive" [$r(57)=0.27$, $p<0.05$] in the speech condition, and with "embarrassed" [$r(57)=0.30$, $p<0.05$], "depressed" [$r(57)=0.34$, $p<0.05$], "apprehensive" [$r(57)=0.28$, $p<0.05$], and "irritated"

Table 1
Mean (S.D.) positive and negative affect responses during task performance in the three experimental conditions

Variable	Reading	Speech	Tracking	*F*
Tense	33.0 (24.2)	42.1 (24.1)	37.8 (26.1)	2.66
Embarrassed	20.3 (17.4)	32.8 (25.8)	13.2 (13.2)	19.85***
Depressed	11.0 (9.9)	17.0 (18.3)	11.1 (11.9)	6.09*
Apprehensive	22.2 (20.9)	30.2 (23.6)	17.3 (18.6)	11.87***
Irritated	10.7 (9.8)	12.0 (11.0)	15.8 (18.6)	3.14*
Involved in the task	62.9 (20.4)	65.6 (21.2)	73.3 (18.5)	8.51***
Satisfied	60.2 (22.8)	45.1 (23.7)	59.9 (22.8)	13.50***
Relaxed	63.4 (22.5)	52.5 (25.0)	55.5 (23.7)	4.65*
Feeling control	65.0 (20.0)	58.5 (22.7)	62.0 (23.8)	2.08
Interested	49.4 (26.6)	51.4 (25.6)	62.5 (28.3)	8.33***

Satisfied=satisfied with own performance; interested=interested in own performance.
n=58.
* $p<0.05$.
*** $p<0.001$.

[$r(57)=0.29, p<0.05$] in the tracking condition. None of the correlations between extraversion and the positive (or negative) affect ratings were significant.

Affect ratings between the tasks (i.e., STPI scores) for the high- and low-neuroticism groups are presented in Fig. 1. Mean anxiety level did not change between the first and the second assessment phases, and there were no significant group differences. However, in the last assessment phase, a highly significant difference between the groups occurred [$T(54)=3.50$, $p=0.001$], resulting in a significant Group×Phase interaction effect [$F(2,53)=4.79$, $p<0.05$]. The main effects of Group and Phase bordered on significance [$F(1,54)=3.72$, $p=0.06$; and $F(2,53)=2.88$, $p=0.07$, respectively]. The high-neuroticism group reported more depressed mood throughout the experiment [main effect of Group: $F(1,54)=9.89$, $p<0.001$]. For both groups, a reduction in depressive mood was observed in the last assessment phase [main effect of Phase: $F(2,53)=3.36$, $p<0.05$]. While the anger ratings were at a minimum for both groups, the main Group effect was still significant [$F(1,55)=4.20$, $p<0.05$]. Neither the main Phase effect nor the Phase×Group interaction effect was significant. In contrast to the scales reflecting negative affect states, no effect of neuroticism on the curiosity ratings was found. However, a small but significant decline in curiosity across the phases was observed [$F(2,53)=5.51$, $p<0.01$]. In conformity with the task-related affect ratings, the relatively high curiosity scores indicate that positive affects prevailed in the subjects in all phases of the experiment. As expected, extraversion was related only to the curiosity scale. Thus, the high-extraversion group reported more curiosity in all assessment phases [main effect of Group: $F(1,54)=4.47$, $p<0.05$].

3.2. Physiological measures

3.2.1. The reading task

Both HR and MAP increased significantly during reading (Table 2) and returned to baseline levels in the recovery phase. None of the main effects or interactions involving Group (i.e., neuroticism and extraversion) were significant. As can be seen in Fig. 2, reading also elicited a highly significant facial LDF increase [main effect of Phase: $F(1,53)=55.78$, $p<0.001$]. The main Group and Group×Phase interaction effects were both insignificant, however. With regard to finger LDF, the continuous increase in response levels seen during task and recovery only approached significance [$F(1,53)=3.45$, $p=0.07$], and no significant Group or Group×Phase interaction effects were found. Using absolute LDF values (expressed in arbitrary units), the increase in facial LDF during reading as compared with baseline values was highly significant [$T(54)=8.70$, $p<0.001$], whereas the difference in finger LDF was insignificant.

3.2.2. The speech task

As shown in Table 2, the speech task induced relatively large increases in both HR and MAP. A more detailed investigation of the response patterns revealed significant HR and MAP increases, even in the preparation phase [$T(55)=8.13$, $p<0.001$, and $T(55)=6.26$, $p<0.001$, respectively], and then a further and larger increase in the speech phase [$T(55)=5.0$, $p<0.001$, and $T(55)=8.94$, $p< 0.001$, respectively]. However, no main effect of Group or Group×Phase interaction effect emerged for any of these physiological variables.

Significant changes in facial LDF occurred across the phases [$F(2,50)=55.45$, $p<0.001$], most notably during the speech phase (Fig. 2). Again, neither the main Group effect nor the Group×Phase interaction effects were significant. Moreover, no effects of

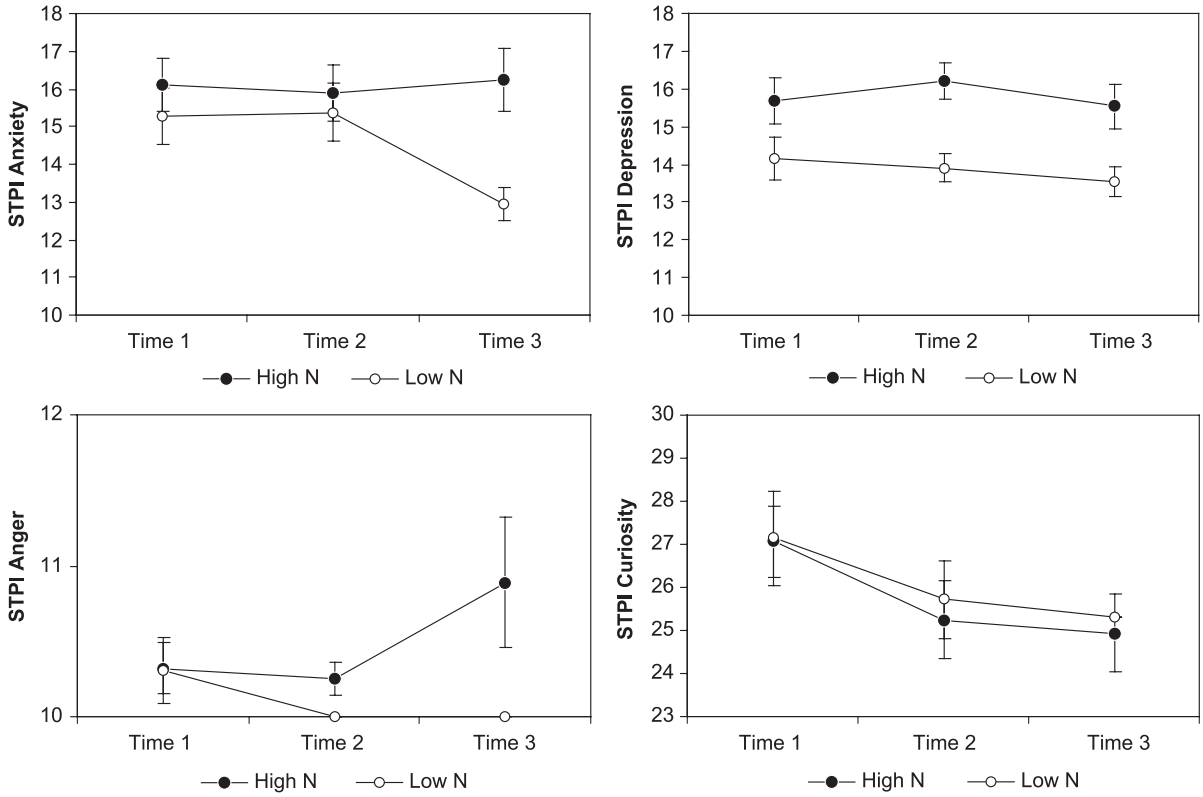

Fig. 1. Mean scores on the STPI scales for high- and low-neuroticism (N) individuals before the tasks (Time 1), after the speech task (Time 2), and at the end of the experiment (Time 3). Error bars represent the S.E.M.

Group, Phase, or their interactions on finger LDF emerged. When absolute LDF values were used, a significant increase in facial LDF [$T(56)=3.51$, $p<0.001$] in the preparation phase and a concomitant decrease in finger LDF [$T(54)=2.61$, $p=0.01$] were demonstrated.

3.2.3. The tracking task

Although the HR and MAP responses were somewhat smaller during the tracking task, as compared with the two other tasks (Table 2), highly significant Phase effects were again demonstrated. During the recovery phase, both HR and MAP reached baseline levels. While a highly significant main phase effect on finger LDF was found [$F(2,52)=9.06$, $p<0.001$], the Phase effect on facial LDF was insignificant. Again, no Group or Group×Phase interaction effect was observed for the LDF variables, or for any of the other physiological variables in this condition. As compared with baseline levels (using absolute values), the increase in facial LDF during tracking and the corresponding decrease in finger LDF were both significant [$T(55)=2.46$, $p<0.05$, and $T(55)=3.62$, $p=0.001$, respectively].

In the correlation analyses, relationships between personality variables (here maintained as continuous variables) and LDF, HR, and MAP response levels during task performance were analyzed for each of the cognitive tasks. However, most of the correlation coefficients were close to zero, and all (i.e., 24 coefficients) were statistically insignificant. Next, relationships between task-related positive and negative affects (i.e., sum scores of ratings for each task) and physiological response levels were investigated. Again, none of the correlation coefficients turned out to be significant. Finally, interaction effects between personality variables and task-related affect level on physiological responding were explored in a series of regression analyses. Models including main effects of the personality and affect variables (one from each set

Table 2
Mean (S.D.) HR and MAP during the three cognitive tasks

Task	HR (S.D.)	*F* (phase effect)	MAP (S.D.)	*F* (phase effect)
Reading task				
Baseline	66.1 (10.7)	67.48***	99.9 (13.1)	72.09***
Reading	75.1 (11.3)		109.7 (14.4)	
Recovery	65.5 (10.7)		101.1 (14.3)	
Speech task				
Baseline	66.0 (10.2)	40.05***	100.8 (15.4)	36.45***
Preparation	72.9 (13.1)		106.3 (16.3)	
Speech	76.8 (12.5)		116.0 (18.1)	
Recovery	64.1 (10.9)		104.1 (16.3)	
Tracking task				
Baseline	65.9 (10.3)	22.35***	102.7 (13.2)	46.61***
Tracking	70.6 (11.5)		107.2 (13.6)	
Recovery	65.5 (10.2)		102.7 (14.6)	

HR=heart rate; MAP=mean arterial pressure; phase=phases within tasks (i.e., baseline, task performance, and recovery).
n=55–56.
*** $p<0.001$.

of variables per analyses), in addition to the interaction term, were tested. However, no significant interaction effects (or main effects) emerged.

3.2.4. Isometric m. masseter contraction

Significant increases in m. masseter EMG levels were observed during reading [$T(55)=5.84$, $p<0.001$], preparation for speech [$T(55)=2.63$, $p<0.05$], speech [$T(55)=5.02$, $p<0.001$], but not during tracking [$T(55)=1.02$, ns.]. No association was found between EMG level or response magnitude during tasks and corresponding facial blood flow changes or other physiological response indices. As shown in Fig. 2, although contraction was associated with a pronounced EMG response [$T(54)=10.15$, $p<0.001$], no change occurred in facial LDF [$T(54)=0.53$, ns.]. A small but significant reduction in finger LDF was observed [$T(54)=2.81$, $p<0.01$].

3.2.5. Additional analyses

Of the four health behavior indices included in the study, only the BMI evidenced a consistent relationship with the cardiovascular measures. Thus, negative correlations (i.e., most coefficients on the order of −0.15 to −0.35) between BMI and HR and/or MAP were found in each of the three conditions. There were also negative correlations between BMI and the LDF measures, but only a couple of the coefficients reached significance. In addition, age was negatively correlated with several of the HR measures. As a consequence of these findings, the analyses of variance were repeated, using BMI and age as covariates. However, no substantial deviation from the first series of analyses was detected. Thus, the main Phase effects were still significant, and no Group or Group×Phase interaction effect emerged for any of the physiological variables.

To elucidate the relationships between general affect *states* during the experimental session and patterns of physiological responding, further analyses were performed, using high/low scores on the STPI scales (averaged across phases) as Group variables. The most consistent finding from these analyses was a relationship between state anxiety and HR, indicating that participants with high STPI anxiety scores exhibited higher HR in one or more phases of each task (data not shown). Thus, in the reading condition, the main effect of anxiety bordered on significance [$F(1,53)=3.83$, $p=0.06$]. In the speech condition, higher HR levels were observed in the high-anxiety group in all phases, and particularly in the preparation phase [Group×Phase interaction effect: $F(3,50)=3.62$, $p<0.05$; main effect of Group: $F(1,52)=3.43$, $p=0.07$]. Finally, in the tracking condition, a significant Group×Phase interaction was again shown, due mainly to a substantial HR elevation in the high-anxiety group in the recovery phase [$F(2,51)=3.63$, $p<0.05$].

4. Discussion

Consistent with results documented in previous studies, increases in facial blood flow as well as arterial pressure and heart rate during task performance were observed in all three tasks. Facial blood flow changes were not related to EMG levels in m. masseter during tasks or during isotonic contraction. Further analyses revealed significant differences in physiological responding across tasks. There was a relatively high MAP level during speech performance and relatively low MAP, HR, and facial LDF levels during tracking. Interestingly, even in the silent preparation phase before the speech, significant

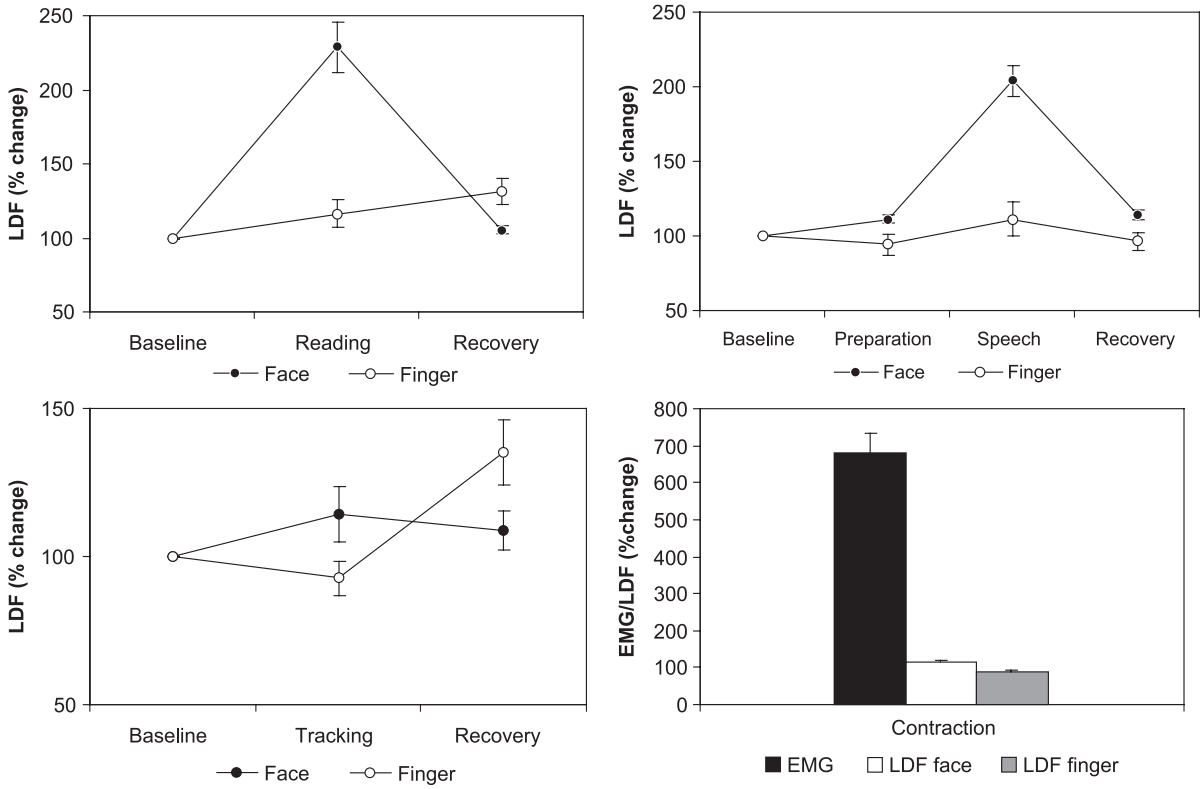

Fig. 2. Facial and finger blood flow changes (LDF) during the cognitive tasks, and EMG and LDF changes during isometric m. masseter contraction. A score of 100 represents response levels at baseline. Error bars represent the S.E.M.

changes in physiological responding were observed. Thus, while the response amplitudes are influenced by the nature of the task, as well as possible sequence effects, taken as a whole, the pattern of cardiovascular responding observed across conditions was remarkably uniform.

The findings also confirmed that task-related negative affect scores were greater in the speech condition than in the other two less personally involving tasks. Thus, the three experimental tasks effectively induced differential response patterns, as evidenced by both cardiovascular and subjective affect indices. The positive affect ratings were consistently higher than the negative affect ratings, particularly in the tracking condition. Thus, although negative emotions were evidently elicited in the subjects, the tasks were also perceived as positive challenges. Moreover, significant positive correlations between neuroticism and task-related negative affect ratings during speech and tracking were found.

Substantial effects of neuroticism on nontask negative affect states measured by the STPI scales were also demonstrated, indicating higher levels of anxiety, depression, and anger in the high neuroticism group. As expected, extraversion was associated only with the STPI curiosity scale. In sum, these findings show that the affective experiences of high- and low-neuroticism individuals were quite different throughout the experiment.

Importantly, however, despite extensive statistical analyses (which increases the likelihood of obtaining accidental significant findings), no clear-cut relationship between personality and facial blood flow, or any other physiological variable, was demonstrated. These results are in agreement with the Drummond (1994) study, which showed that changes in facial blood flow, measured with photoplethysmography, during mental arithmetic was unrelated to neuroticism or extraversion. In contrast to Drummond's study, the present investigation used laser Doppler flowmetry,

and facial blood flow changes were measured in three experimental conditions, which induced different cardiovascular response patterns.

There are several possible explanations for these negative findings. First, it can be argued that individual differences in neuroticism and extraversion will be related to psychophysiological differences only to the extent that they interact with situational conditions to produce relatively *strong emotional arousal*. The intensity of the eliciting situation or task is obviously important in effecting this process (Geen, 1997). It can also be argued that only extreme scorers on the personality dimensions in question may exhibit strong responses to the experimental tasks. While facial flushing and blushing are undoubtedly elicited by intense emotional experiences in certain everyday life circumstances, such as embarrassing interpersonal situations, ethical considerations set a limit on what can be done in an experimental setting. Thus, the relatively weak and trivial stimuli that are typically employed in laboratory studies cannot be expected to elicit the sort of physiological response associated with strong emotions, such as rage or high levels of anxiety and embarrassment. In consequence, moderator effects of personality can be expected only in particular circumstances. It should be emphasized, however, that the speech task evidently induced a relatively strong stress response in the subjects, as judged by the cardiovascular data in particular.

Second, facial blood flow responses seem to be triggered by a fairly broad range of stimuli, both physiological and psychological, which may compete with influences from the affect response system. Moreover, common cognitive tasks seem to provoke a mixture of positive and negative emotions that may be related to facial blood flow changes in different ways. In addition, both body heating (e.g., Drummond and Finch, 1989; Drummond and Lance, 1987; Nordin, 1990) and pain (e.g., Kemppainen et al., 2001) evoke facial blood flow changes. For example, the study by Kemppainen et al. (2001) showed that high-intensity tooth pulp stimulation caused a transient blood flow increase in the cheek and forehead, together with a transient blood flow reduction in the skin of the nose. Taken together, these studies indicate, first, that various psychological and sensory stimuli may trigger facial blood flow changes, and second, that separate vasoactive reflex mechanisms may exist for different orofacial regions. Thus, blood flow data may represent the *sum of activity* in several regulatory mechanisms.

A third possibility is that the facial blood flow response is an *integrated part* of a larger physiological and behavioral response pattern elicited during stress. The periaquaductal gray area (PAG) appears to be one of the structures that integrate responses to stress, threat, and pain (Bandler and Shipley, 1994; Clement et al., 2000). The PAG is involved in at least three integrated response patterns. The dorsolateral PAG controls a pattern observed during active responding, consisting of elevated arterial pressure and heart rate, increase in blood flow to the skeletal muscles, behavioral reactivity, and nonopioid hypoalgesia. A second area connected to the first is involved in facial blood flow and seems to be of importance in integrating responding during social encounters. The ventrolateral part of the PAG controls a pattern consisting of behavioral inhibition, depressor responses, bradycardia, and opioid hypoalgesia. According to this model, both active responding (active coping) and inhibition of responding are linked to distinct hypoalgesic mechanisms and cardiovascular response patterns, also including facial blood flow responses.

Possibly then, facial blood flow responses may be likened to other psychophysiological response systems, as far as the generality of the response is concerned. The electrodermal response system, for example, has been linked to a number of psychological states and processes (Dawson et al., 2000). Moreover, changes in electrodermal activity, like facial blood flow changes, do not occur in isolation but as part of a complex of responses mediated by the autonomic nervous system. For example, Pennebaker et al. (1987) demonstrated that, in a student sample, talking about traumatic events was associated with decreased finger skin conductance levels, together with increased cardiovascular activity. In accordance with Gray's (1987) theory of anxiety and Fowles' (1980) extension of Gray's work, this response pattern was interpreted as indicative of decreased behavioral inhibition. Fowles (1980) has shown convincingly that behavioral inhibition is specifically linked to skin conductance activity, whereas behavioral activation is associated with increased cardiovascular action.

Similarly, in the present study, a dissociation between responses in the digital and facial vasculature was repeatedly observed during the experiment. Thus, increases in facial blood flow during the preparation phase before the speech task as well as during the tracking task occurred in parallel with decreases in finger blood flow. Furthermore, while facial blood flow returned to baseline levels in the recovery period after reading, finger blood flow showed a significant increase. Such data suggest that psychophysiological skin blood flow responses, analogous with skin conductance responses, are more strongly influenced by the overall response (coping) pattern elicited by a specific stimulus or task than by broad personality dispositions.

In general, research has essentially failed to uncover any consistent psychophysiological differences between those high and low in neuroticism or trait anxiety (Fahrenberg, 1992; Fahrenberg and Foerster, 1982; Geen, 1997). On the other hand, heightened state anxiety during challenge is associated with a propensity for increased cardiovascular responsiveness and particularly when the individual can exert some control over the stressor (referred to as active coping, see Obrist et al., 1978). In agreement with this generalization, the present study showed that individuals with heightened state anxiety also exhibited the highest HR responses during the experiment. State anxiety was not related to facial or digital blood flow changes, however.

A couple of additional issues should be briefly commented on. Despite the common sense belief that the face sometimes becomes pallid or livid during intense emotions such as rage (Darwin, 1872/1965), to our knowledge, such pronounced facial blood flow decreases have not been documented in experimental studies. Thus, a first question is under what psychological conditions the face is expected to go pale. Early studies by Ax (1953) and Schachter (1957) showed that facial temperature (which correlates moderately with facial blood flow) sometimes fell when subjects were rudely abused by the experimenter or were manipulated to believe that they were in danger of electrocution. However, these findings are difficult to verify on ethical grounds. Taken together, the findings, so far, are consistent with the notion that the face flushes during moderate psychological challenges and frustrations. However, the face may turn pale during more extreme emotions such as rage and strong anxiety, possibly because facial flushing is overruled by strong sympathetic cutaneous vasoconstriction as part of the fight–flight response (Drummond, 1997a). Maybe, experimental conditions likely to induce variants of passive coping, and a decrease in arterial pressure (e.g., injections or venepuncture) would also induce a decrease in facial blood flow.

Second, an interesting hypothesis in need of investigation is whether the phenomenon of individual response stereotypy (Lacey and Lacey, 1958; Wenger and Cullen, 1972) may also include facial blood flow responses. Generally, we need to know more about the consistency and possible functional implications of flushing. For example, it has been hypothesized that perceived changes in facial temperature associated with facial blood flow may influence emotional experience (Zajonc et al., 1989). In addition, it has been suggested that facial blood flow may have a role in emotional expression by signaling the individual's emotional state in social interaction (e.g., Drummond, 1999). Incidentally, both these hypotheses are of obvious relevance for the understanding of mechanisms involved in facial flushing and blushing associated with embarrassment or shame (e.g., de Jong et al., 2003; Drummond et al., 2003).

In conclusion, this study demonstrates that brief verbal and nonverbal tasks can induce facial blood flow elevations to varying degrees. In particular conditions, the facial vasodilatation response is accompanied by a decrease in finger blood flow. No modifying influences of personality or self-reported affect states on blood flow responses were found. These findings accord with accumulated research, showing that neuroticism and extraversion are, at best, only weakly related to state and reactivity parameters of psychophysiological activation processes. Thus, the facial blood flow changes observed during commonly used laboratory tasks are probably determined by factors other than personality variables, e.g., the nature of the problem or task at hand, as well as the coping strategy adopted by the subject. To explore this hypothesis, facial blood flow needs to be studied across a spectrum of stimuli, tasks and coping behaviors.

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